

Mariam Ahmed

MECHATRONICS ENGINEERING STUDENT | EMBEDDED SYSTEMS &
ROBOTICS ENTHUSIAST

+0201011178117 | mariamahmedkamal2005@gmail.com | 📍 Cairo, Egypt | [Mariam Ahmed](#)

Objective

Mechatronics Engineering student passionate about embedded systems, robotics, and intelligent automation. Skilled in C/C++, Arduino, and developing practical engineering solutions. Seeking an internship to apply technical and problem-solving skills in real-world projects.

Education

Bachelor of Mechatronics Engineering — Egyptian Chinese University (ECU)

Junior Student | GPA: 3.8 / 4.0

Expected Graduation: August 2028

Projects

Autonomous Medical Delivery Robot

- Developing an autonomous medical delivery robot integrated with a Flutter mobile application and Firebase Realtime Database for real-time hospital delivery management.
- Implementing RFID authentication, ESP32 communication, and AI-based automation for secure medicine delivery and smart healthcare assistance.

Charging Station

- Developed a smart hybrid charging station integrating solar energy and grid power for safe and efficient battery charging.
- Implemented ESP32-based real-time monitoring and control system with voltage, current, and temperature sensors for multi-slot battery management.

Delta Robot with Computer Vision – Pick & Place Automation System

- Developed a Delta robot for automated pick-and-place operations.
- Integrated computer vision for object detection and sorting.
- Applied motion control and kinematic design concepts.

Smart Home Automation System

- Designed an automated system to control lighting and fan speed based on temperature.
- Integrated LM35, LDR, and flame sensor with buzzer alerts.
- Used ATmega32 with ADC, PWM, and GPIO drivers.

Vehicle Fault Detection System

- Designed dual ECU system communicating via UART.
- Monitored faults using sensors, EEPROM, and LCD.

Courses

Embedded Systems Diploma – Edges for Training (June2025 - Sep2025)

- Studied Embedded C, AVR interfacing, RTOS basics.
- Completed multiple hardware-based projects.

Embedded Systems Workshop – Egyptian Chinese University (June2025-July2025)

- Gained hands-on experience with microcontrollers and basic interfacing.

Mechanical Design Workshop – Pascal Academy (Jan2025-Feb2025)

- Learned basics of mechanical design and engineering drawing.

Training & Activities

Technical Field Visit – Cairo Metro (August 2025)

- Observed maintenance of electrical and mechanical systems.
- Learned about control systems, power, and safety procedures.

Technical Skills

Embedded Systems Development

Hands-on experience in programming and developing embedded applications using Arduino and AVR microcontrollers with C/C++.

Microcontroller Programming & Interfacing

Skilled in working with ATmega32, GPIO, UART, and basic communication between hardware components and embedded systems.

Programming Languages

Strong command of C and C++ with good understanding of data structures and problem-solving concepts.

RTOS & Embedded Concepts

Familiar with RTOS basics, task management concepts, and embedded software architecture fundamentals.

Mobile & Software Development

Basic experience in Flutter development and building simple cross-platform applications.

Database Management

Good knowledge of SQL and SQLite for data storage and basic database operations.

Mechanical Design (SolidWorks)

Experience in 3D modeling, assemblies, and mechanical design for mechatronics applications using SolidWorks.

Soft Skills

- Strong problem-solving and analytical thinking skills
- Leadership mindset with responsibility ownership
- Effective teamwork and collaboration abilities
- Good communication and technical presentation skills
- Strong time management and ability to meet deadlines
- Fast learner with high adaptability to new environments
- Creative thinking and innovative approach to challenges
- Bilingual: Arabic & English